

- 4 (a) Fig. 4.1 shows two identical small positive charged spheres A and B, each having a mass of $5.0 \times 10^{-4} \text{ kg}$ in equilibrium. Sphere A is fixed in position by an insulated rod while sphere B is free to move and attached to the ceiling by a string with length 0.20 m.

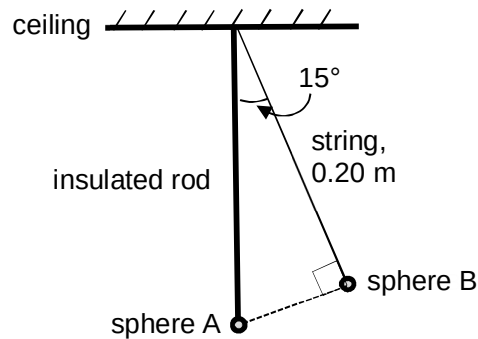


Fig. 4.1

The string suspends at an angle of 15° to the vertical insulating rod and it also makes a right angle with the straight line connecting the two spheres.

Calculate

- (i) the repulsive force acting on sphere B due to sphere A;

(ii) the magnitude of the charge on each sphere.

charge = C [2]

- (b) Sphere B connected with the same string as part (a) is now projected into the page at a speed of 0.50 m s^{-1} such that it orbits horizontally around the same vertical insulating rod as shown in Fig. 4.2.

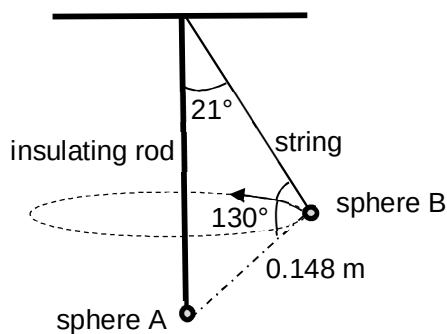


Fig. 4.2

The string makes an angle of 21° with the insulating rod. The distance between sphere A and sphere B is now found to be 0.148 m and the angle between the string and this line connecting sphere A and B is now 130° .

- (i) Show that the resultant horizontal force on sphere B is $1.7 \times 10^{-3} \text{ N}$.

(ii) Calculate the tension in the string.

tension = N [3]

(c) A uniform magnetic field perpendicular to the motion of sphere B is now introduced to the system in part (b) as shown in Fig. 4.3.

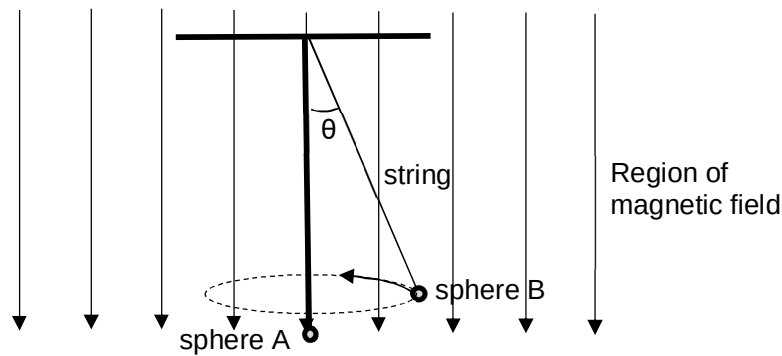


Fig. 4.3

Discuss how this might affect the subsequent motion of sphere B.

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[3]

- (d)** In the presence of air, sphere A and B will slowly discharge as they ionize the air particles around them.

Describe the path of sphere B when this happens.

[1]